FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number		N_a	$= 6.023 \times 10^{23}$
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):

(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

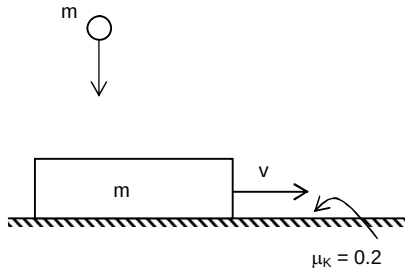
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A ball of mass m falls vertically from a height h and collides with a block of equal mass m moving horizontally with a velocity v on a surface. The coefficient of kinetic friction between the block and the surface is $\mu_k = 0.2$, while the coefficient of restitution (e) between the ball and the block is 0.5. There is no friction acting between the ball and the block. The velocity of the block just after the collision decreases by (if it is still in motion)



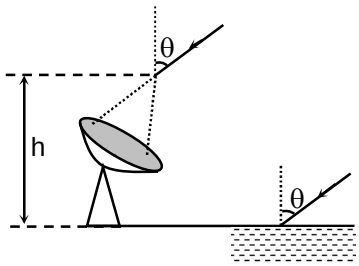
(A) $0.5\sqrt{2gh}$

(B) 0

(C) $0.1\sqrt{2gh}$

(D) $0.3\sqrt{2gh}$

2. Radio waves of wavelength λ at an angle θ to vertical are received by a radar after reflecting from a nearby water surface and directly. If the radar records a maximum intensity, the height of antenna *h* from water surface can be

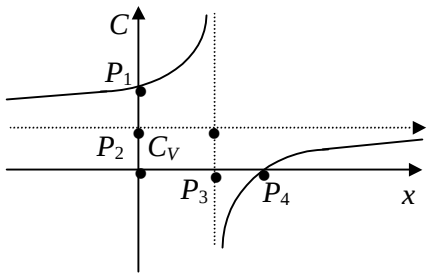


(A) $\frac{\lambda}{2\cos\theta}$ (B) $\frac{\lambda}{2\sin\theta}$

(C) $\frac{\lambda}{4\sin\theta}$ (D) $\frac{\lambda}{4\cos\theta}$

Space for Rough work

3. The following graph gives the variation of molar heat capacity C a function of x where x is given by the expression PV^x = constant. If gas is monoatomic what are the coordinates of P_1 and P_4



(A) $\left(0, \frac{5}{2}R\right)$ & $\left(\frac{5}{3}, 0\right)$

(B) $\left(0, \frac{3}{2}R\right)$ & $\left(\frac{7}{5}, 0\right)$

(C) $\left(0, \frac{5}{2}R\right)$ & $\left(\frac{7}{5}, 0\right)$

(D) $\left(0, \frac{7}{5}R\right)$ & $\left(\frac{5}{3}, 0\right)$

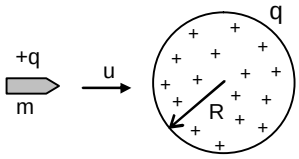
(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. A closed organ pipe of length 28 cm closed at one end is found to be at resonance when a tuning fork of frequency 850 Hz is sounded near the open end. If velocity of sound in air is 340 m/s, then the
   (A) air in the pipe is vibrating in fundamental mode
   (B) air in the pipe is vibrating in first overtone
   (C) end correction of the pipe is 1 cm
   (D) end correction of the pipe is 2 cm

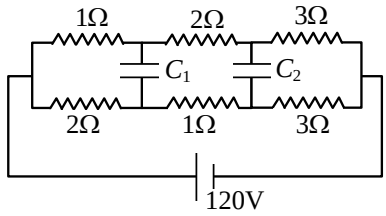
Space for Rough work

5. Which of these options are correct?



(A) A bullet of mass m and charge q is fired towards a solid uniformly charged sphere of radius R and total charge +q. If it strikes the surface of sphere with speed u. The minimum speed u so that it can penetrate through the sphere. (Neglect all resistance forces or friction acting on bullet except electrostatic forces) is $\frac{q}{\sqrt{4\pi\varepsilon_0 mR}}$

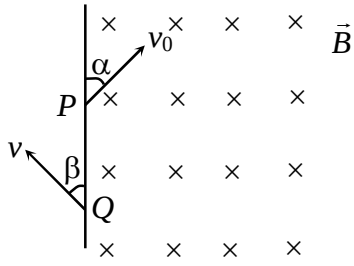
(B) In the above diagram the minimum speed u so that it can penetrate through the sphere. (Neglect all resistance forces or friction acting on bullet except electrostatic forces) is $\frac{q}{\sqrt{2\pi\varepsilon_0 mR}}$



(C) In the circuit shown in figure $C_1 = C_2 = 2\mu F$. Then charge stored in capacitor C_2 is zero

(D) In the above circuit charged stored in capacitor C_1 is 40 μC.

6. A particle of charge $-q$ and mass m enters a uniform magnetic field $\vec{B}$ (perpendicular to paper inwards) at P with a velocity v_0 at an angle α and leaves the field at Q with velocity v at angle β as shown in the figure. Then



(A) $\alpha = \beta$

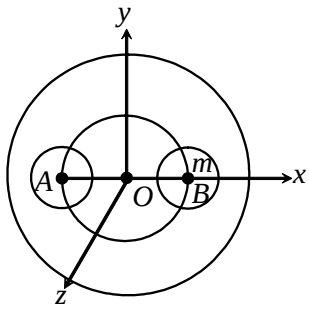
(B) $v = v_0$

(C) $PQ = \frac{2mv_0 \sin\alpha}{Bq}$

(D) particle remains in the field for time $t = \frac{2m(\pi - \alpha)}{Bq}$

Space for Rough work

7. A solid sphere of uniform density and radius 4 units is located with its centre at the origin *O* of co-ordinates. Two spheres of equal radii 1 units, with their centres at *A*(–2,0,0) and *B* (2, 0 ,0) respectively, are taken out of the solid leaving behind spherical cavities as shown in figure. Then



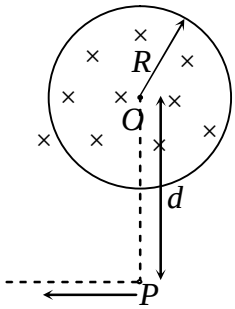
(A) the gravitational field due to this object at the origin is zero.

(B) the gravitational field at the point *B*(2, 0, 0)is zero

(C) the gravitational potential is same at all points on the circle $y^2 + z^2 = 36$

(D) the gravitational potential is same at all points on the circle $y^2 + z^2 = 4$

8. In a cylindrical region of radius *R*, there exists a time varying magnetic field *B* such that $\frac{dB}{dt} = k(>0)$. A charged particle having charge q is placed at the point *P* at a distance *d* (> R) from its centre O. Now, the particle is moved in the direction perpendicular to OP (see figure) by an external agent upto infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s.



(A) Work done by external agent is $\frac{q\pi R^2}{4}k$ if d = 2R

(B) Work done by external agent is $\frac{q\pi R^2}{8}k$ if d = 4R

(C) Work done by external agent is $\frac{q\pi R^2}{4}k$ if d = 4R

(D) Work done by external agent is $\frac{q\pi R^2}{4}k$ if d = 6R

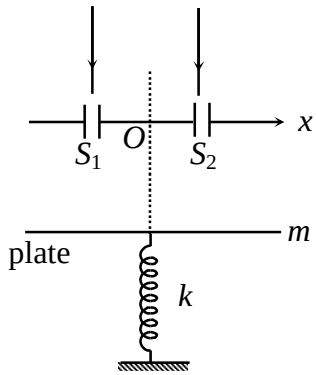
Space for Rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 9 and 10

Two slits S_1 and S_2 lie on the x-axis and symmetric with respect to y-axis are illuminated by a parallel monochromatic light beam of wavelength λ as shown. The distance between slits is d (>> λ). Point O is the mid point of the line S_1S_2 and this point is considered as the origin. The slits are in horizontal plane. The interference pattern is observed on a horizontal plate (acting as screen) of mass m which is connected to one end of a vertical massless spring of spring constant k. The other end of the spring fixed to ground. At t = 0, the plate is at a distance D (>> d) below the plane of slits and spring is in its natural length. The plate is released from rest from its initial position.



9. The rate by which fringe, width will increase when acceleration of plate is zero, is

(A) $\frac{\lambda g}{d}\sqrt{\frac{m}{k}}$ (B) $\frac{\lambda g}{3d}\sqrt{\frac{m}{k}}$

(C) $\frac{\lambda g}{4d}\sqrt{\frac{m}{k}}$ (D) $\frac{\lambda g}{2d}\sqrt{\frac{m}{k}}$

10. A thin slab of refractive index μ is kept in front of one of slits such that position of first maxima shift to the position of central maxima at the instant when plate has been held at rest initially. The thickness of slab is

(A) $\frac{d}{(\mu-1)}$ (B) $\frac{d\lambda}{D(\mu-1)}$

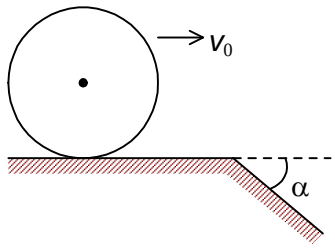
(C) $\frac{D\lambda}{d(\mu-1)}$ (D) $\frac{\lambda}{\mu-1}$

Space for Rough work

SECTION – C
(Single digit integer type)

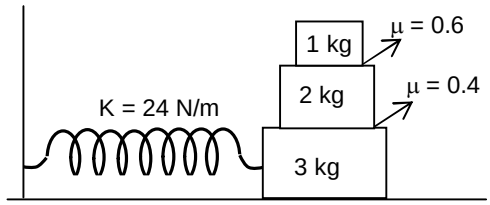
This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

11. A uniform solid cylinder of radius, R = 15 cm rolls over a horizontal plane passing into an inclined plane forming an angle α = 30° with the horizontal as in Figure. Find the maximum integral value of the velocity v_0 (in m/s) which still permits the cylinder to roll on to the inclined plane section without a jump. The sliding is assumed to be absent.



12. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc. Find the torque (in N-m) acting on the disc after 15 sec. (Take $4kQ$ = 1 S.I. unit and R = 1m)

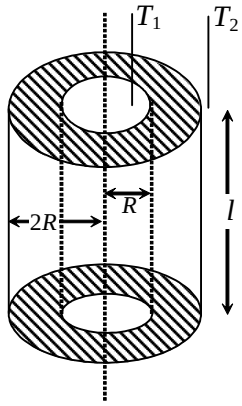
13. What can be the maximum amplitude (in m) of the system so that there is no slipping between any of the blocks



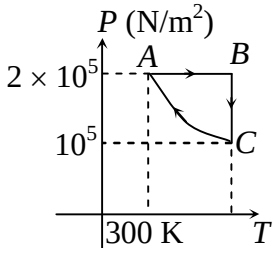
14. A hydrogen-like atom of atomic number Z is in an excited state of quantum number 2n. It can emit a maximum energy photon of 204 eV. If it makes a transition to quantum state n, a photon of energy 40.8 eV is emitted. Ground state energy of hydrogen atom is -13.6 eV. Find the value of n + Z.

Space for Rough work

15. Inner surface of a cylindrical shell of length l and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that $(T_1 > T_2)$ as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and 2R respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{l}{2}$, inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk. If rate of radial outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2, then find the value of n.

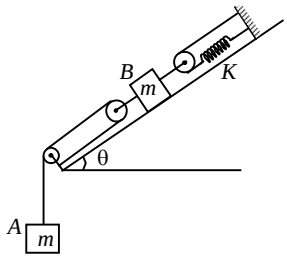


16. Two moles of a monatomic ideal gas is taken through a cyclic process shown on pressure (P)-temperature (T) diagram in figure. Process CA is represented as PT = Constant. If efficiency of given cyclic process is $1-\frac{3x}{12\ln 2+15}$, then find the value of 'x'.



17. A bus is moving towards a huge wall with a velocity of 5 m/s. the driver sounds a horn of frequency 200 Hz. What is the frequency of beats heard by a passenger of the bus (to the nearest integer), if the speed of sound in air is 330 m/s.

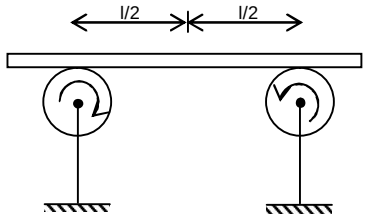
18. Two blocks *A* and *B*, each of mass *m* are connected by means of a pulley-spring system on a smooth inclined plane of inclination θ as shown in the figure. All the pulleys and spring are ideal. Now, *B* is slightly displaced from its equilibrium position. It starts to oscillate. Time period of oscillation of *B* will be $n\pi$ seconds. Find the value of 'n'. (Take *m* = 4 kg, *K* = 5 N/m, π = 3.14)



Space for Rough work

19. A drop of water of mass m = 0.4 g is placed between two clean glass plates, the distance between which is 0.01 cm. The force of attraction between the plates is F. Surface tension of water = 0.08 N/m and density of order = 1000 kg/m^3. Find the value of [F], where [F] is greatest integer function of 'F'.

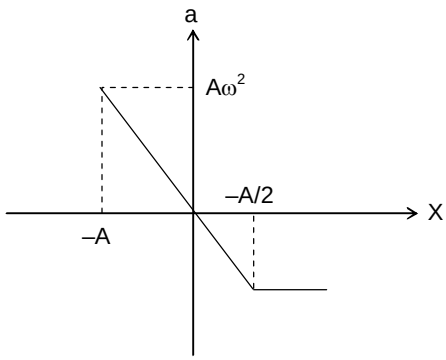
20. A uniform rod is placed on two spinning wheels as shown in the figure. The axes of the wheels are separated by l. The coefficient of friction between the rod and the wheel is μ. If the period of small oscillations is $\pi\sqrt{\frac{a\ l}{2\mu g}}$, a=



SECTION – D
(Numerical Based XXXXX.XX answer Type)

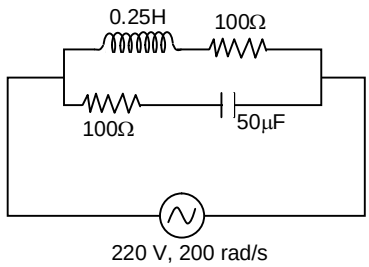
This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

21. Following is the acceleration vs position graph of a particle performing oscillatory motion along x-axis. ω = 2 rad/s. What is the time period of the given motion? (Take $\sqrt{3}$ = 1.73 and π = 3.00)



22. A train is moving along +ve x-axis with 20 m/s and its engine is located at (–400 m, 0) at an instant. It is sounding horn of frequency 600 Hz. A person is listening to the horn standing at (0, –300 m). If wind is blowing at 10 m/s towards +ve x-axis, then what is the apparent frequency heard by person? Speed of sound in air is 330 m/s.

23. In the given Ac circuit, what is the power developed in watts?



Space for Rough work

Chemistry PART – II

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

24. How many maximum number of electrons of an atom will have the following set of quantum numbers?

n = 4

ℓ = 0, 1, 2

m = 0, $\pm$1

s = $\pm$½

(A) 18 (B) 12

(C) 14 (D) 16

25. The half-life of a chemical reaction is 20s, when the reaction starts with 0.4 M concentration of the reactant. The half-life remains the same when the reaction starts with 0.8 M concentration of the reactant. What is the unit of rate constant of the reaction?

(A) mol $L^{-1}s^{-1}$ (B) s^{-1}

(C) mol^{-1} L^2s^{-1} (D) mol L^2s^{-1}

26. $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$

The correct statement regarding the above reaction is(Assume that only NO_2 is initially taken in the reaction container).

(A) 'K_C' changes by changing the concentrations of the species NO_2, NO and O_2.

(B) [NO] α [O_2]

(C) The reaction proceeds to backward direction by decreasing pressure at equilibrium

(D) Inert gas has no effect at constant pressure

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

27. $MnO_4^- + H^+ + C_2O_4^{2-} \longrightarrow Mn^{2+} + CO_2 + H_2O$
Which of the following can be the stoichiometric coefficient(s) of the species involved in the above reaction, after balancing it?
(A) 2 (B) 10
(C) 8 (D) 6

28. Which of the following compound(s) can form buffer solution(s), when added to calculated amount of NaOH?
(A) CH_3COOH (B) HCN
(C) H_2SO_4 (D) HNO_3

29. Which of the following contain(s) unpaired electrons?
(A) Oxygen atom (B) Dioxygen molecule
(C) Nitric oxide (D) Dinitrogen molecule

30. Which of the following compounds can form silicates with SiO_2?
(A) NaOH (B) Na_2CO_3
(C) NaCl (D) $NaNO_3$

31. Which of the following exists as polymer(s)?
(A) B_2H_6 (B) BN
(C) SiH_4 (D) H_3BO_3

Space for Rough work

(Paragraph Type)

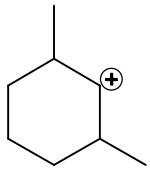
This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

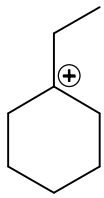
Paragraph for Question Nos. 32 and 33

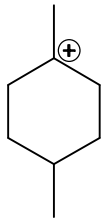
Carbocations are formed as reaction intermediates in many organic reactions. They are stabilized by resonance, hyperconjugation and inductive effects. A less stable carbocation can undergo rearrangement to a more stable carbocation by hydrogen or methyl shifts. They are formed by heterolytic fission of covalent bonds assisted by heat or polar solvents.

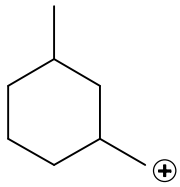
Answer the following questions on the basis of above write up.

32. Which of the following carbocation will be most stable if only hyperconjugation is considered as the deciding factor?

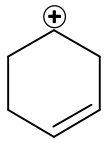
(A) 

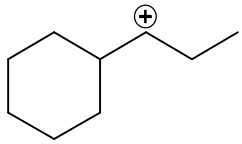
(B) 

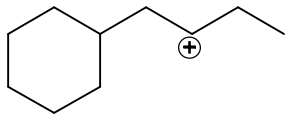
(C) 

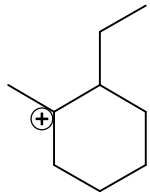
(D) 

33. Which of the following carbocation does not undergo rearrangement?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

34. The colour of an inorganic compound(X), containing four elements is orange. On heating it forms a green solid(Y), a colourless gas(Z) and a liquid which pH at room temperature is seven. What is the sum of the total number of atom(s) present in (Y) and (Z)?

35. The normal boiling point of water increases by 4.16^oC, if two moles of a salt is added to one kg water. What is the van't Hoff factor of the salt? [K_b of water = 0.52 K kg mol^{-1}]

36. How much heat is absorbed in Kcal mol^{-1} unit by heating one mole of an ideal gas from 300 K and 1300 K at constant volume? $\left[C_V = \frac{3}{2}R\right]$

37. $Zn(s) \mid Zn^{2+}(0.01\ M) \parallel H^+ \mid H_2(1\ atm), Pt$
The e.m.f of the above electrochemical cell is 0.6418 V. What is the pH of the cathode half-cell if $E^o_{Zn^{2+}/Zn} = -0.76\,V$

38. $NH_4NO_3 \longrightarrow N_2O + 2H_2O$
Let the molecular mass of NH_4NO_3 is M, What will be the value of 'n', if the equivalent mass of NH_4NO_3 is expressed as $\frac{M}{n}$?

39. Two containers contain equal moles of SO_2 and CH_4 gases separately at constant temperature and pressure. If the simplest ratio of their relative rates of effusion is expressed as x : y, the value of (x + y) is:
[Assume same area of orifice on the containers]

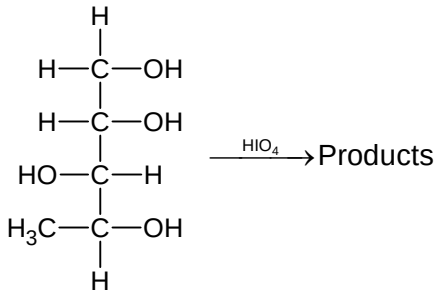
40. How many complete octahedral void(s) is/are occupied by Na^+ ions in NaCl unit cell?

Space for Rough work

41. $Cr_2(SO_4)_3 \xrightarrow[(ii)\ Na_2O_2]{(i)\ NaOH} \text{Yellow solution}$

If the oxidation number of chromium in the compound that provides yellow colour is +x, then 'x' is

42. What is the sum of the number of linkage isomers and ionization isomers possible for the octahedral complex with formula $Cr(NH_3)_4(CN)(Br)(Cl)$?

43. 

How many different type of product(s) in the above reaction can give Tollen's test?

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

44. $X(g) \rightleftharpoons Y(g) + Z(g)$

250 mole of 'X' is taken in a 100 L container at a certain temperature. 50 mole percentage of 'X' undergoes reaction at equilibrium. How many moles of 'X' should be added at equilibrium so that concentration of Y will become 1.3 M at the new equilibrium?

45. The pH of 0.1 M aqueous solution of the sodium salt of a weak monobasic acid(HA) is 12.8. What is the p^{K_a} value of the acid?

46. Two moles of glycol is dissolved in 891.24 g of water. The solution is cooled to -4.8^oC. How many gram of ice is formed in the cooling process? [K_f of water = 1.86 K kg mol^{-1}]

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

47. If $\vec{r}.\vec{a}=0, \vec{r}.\vec{b}=1$ and $\left[\vec{r}\ \ \vec{a}\ \ \vec{b}\right]=1, \vec{a}.\vec{b}\neq 0, \left(\vec{a}.\vec{b}\right)^2-\left|\vec{a}\right|^2\left|\vec{b}\right|^2=1$ the value of $\vec{r}$ in terms of $\vec{a}$ and $\vec{b}$ is

(A) $\vec{a}\times\left(\vec{a}\times\vec{b}\right)+\vec{a}\times\vec{b}$ (B) $\vec{a}\times\left(\vec{a}\times\vec{b}\right)+\dfrac{\vec{a}\times\vec{b}}{\left|\vec{a}\times\vec{b}\right|^2}$

(C) $\vec{a}\times\left(\vec{b}\times\vec{a}\right)+\dfrac{\vec{b}\times\vec{a}}{\left|\vec{b}\times\vec{a}\right|^2}$ (D) $\dfrac{\vec{a}\times\left(\vec{a}\times\vec{b}\right)}{\left|\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right|}+\vec{a}\times\vec{b}$

48. The solution of the differential equation $\dfrac{d^2y}{dx^2}=\sin 3x+e^x+x^2$ when $y_1(0)=1$ and $y(0)=0$ is

(A) $\dfrac{-\sin 3x}{9}+e^x+\dfrac{x^4}{12}+\dfrac{1}{3}x-1$ (B) $\dfrac{-\sin 3x}{9}+e^x+\dfrac{x^4}{12}+\dfrac{1}{3}x$

(C) $\dfrac{-\cos 3x}{3}+e^x+\dfrac{x^4}{12}+\dfrac{1}{3}x+1$ (D) none of these

49. The solution set of $\log_{|\sin x|}\left(x^2-8x+23\right)>\dfrac{3}{\log_2|\sin x|}$ contains

(A) $x\in(3,\pi)\cup\left(\pi,\dfrac{3\pi}{2}\right)\cup\left(\dfrac{3\pi}{2},5\right)$ (B) $x\in(3,\pi)\cup(\pi,5)$

(C) $x\in\left(3,\dfrac{5\pi}{2}\right)$ (D) none of these

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. The real values of a, b, p, q for which $(2x-1)^{20}-(ax+b)^{20}=(x^2+px+q)^{10}$ are

(A) $p=q=\frac{1}{2}$ (B) $b=-2a=\pm\left(2^{20}-1\right)^{1/20}$

(C) $2b=-a=\pm\left(2^{20}-1\right)^{1/20}$ (D) $4q=-p=1$

51. If $|z_1|=|z_2|=|z_3|=1$ and $S=|z_2-z_3|^2+|z_3-z_1|^2+|z_1-z_2|^2$. Let $\alpha=$ minimum value of S and $\beta=$ maximum value of S as z_1,z_2,z_3 vary on $|z|=1$, then

(A) $\alpha=0$ (B) $\beta=9$

(C) $\beta=3$ (D) $\alpha=2$

52. A four digit number (numbered from 0000 to 9999) is said to be lucky if the sum of first two digits is equal to the sum of its last two digits. If a four digit number is picked up at random, then the probability that it is lucky is

(A) >0.065 (B) <0.068

(C) 0.066 (D) 0.067

53. Let $\alpha(n)=1-\frac{1}{2}+\frac{1}{3}-.....+(-1)^{n-1}\frac{1}{n}$

(A) $\frac{1}{n+1}+\frac{1}{n+2}+.....+\frac{1}{2n}=\alpha(2n)$ (B) $\alpha(2n)<1\forall n$

(C) $\alpha(2n)\geq 0.5\,\forall n$ (D) $0.5<\alpha(n)<1\forall n$

54. Let $f(x)=ax^2+bx+c$, where $a,b,c\in R$. Suppose $|f(x)|\leq 1\forall x\in[0,1]$, then

(A) $|a|\leq 8$ (B) $|b|\leq 8$

(C) $|c|\leq 1$ (D) $|a|+|b|+|c|\leq 17$

Space for Rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 55 and 56

$\alpha, \beta, \gamma, \delta$ are angles in I, II, III and IV quadrant respectively and no one of them is an integral multiple of $\frac{\pi}{2}$. They form an increasing arithmetic progression.

55. Which of the following is correct?

(A) $\cos(\alpha+\delta)>0$ (B) $\sin(\alpha+\delta)=0$

(C) $\cos(\alpha+\delta)<0$ (D) $\cos(\alpha+\delta)>0$ or $\cos(\alpha+\delta)<0$

56. If $\alpha+\beta+\gamma+\delta=\theta$ and $\alpha=70^o$, then

(A) $400^o<\theta<580^o$ (B) $470^o<\theta<650^o$

(C) $680^o<\theta<860^o$ (D) $540^o<\theta<900^o$

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

57. Let $f(x)=\begin{vmatrix} 2\cos^2 x & \sin(2x) & -\sin x \\ \sin(2x) & 2\sin^2 x & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$. Find $\frac{1}{\pi}\int_0^{\pi/2}\left[f(x)+f'(x)\right]dx$.

58. Let $A=\begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$ and $(A+I)^{50}-50A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, find $a+b+c+d$.

59. If $f(n)=\sum_{k=1}^{n}\sum_{j=k}^{n}\left({}^nC_j\right)\left({}^jC_k\right)$ find $f(6)-660$

Space for Rough work

60. Tangent are drawn from any point on the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$ to the circle $x^2+y^2=9$. If the locus of the mid – point of the chord of contact is $a\left(x^2+y^2\right)^2=bx^2-cy^2$, then the value of $a^2+b^2+c^2-7870$ is equal to

61. Let $0\le\beta_r\le1$ and $\sum_{r=1}^{k}\cos^{-1}\beta_r=\frac{k\pi}{2}$ for any $k\ge1$ and $A=\sum_{r=1}^{k}(\beta_r)^r$, then $\lim\limits_{x\to A}498.\frac{\left(1+x^2\right)^{1/3}-(1-2x)^{1/4}}{x+x^2}$ is equal to λ, where $\frac{\lambda}{83}=$

62. For the circle $x^2+y^2=r^2$, the value of r for which the area enclosed by the tangents drawn from the point (6, 8) to the circle and the chord of contact is maximum is

63. If $\int\frac{2e^{5x}+e^{4x}-4e^{3x}+4e^{2x}+2e^x}{\left(e^{2x}+4\right)\left(e^{2x}-1\right)^2}dx=\tan^{-1}\left(e^{x/2}\right)-\frac{K}{248\left(e^{2x}-1\right)}+C$ then K – 120 is equal to

64. If Q is the foot of the perpendicular from the point P (4, –5, 3) on the line $\frac{x-5}{3}=\frac{y+2}{-4}=\frac{z-6}{5}$ then $25(PQ)^2-450=2$

65. If $y=\left(1+\frac{1}{x}\right)^x$ then $\frac{2\sqrt{y_2(2)+\frac{1}{8}}}{\left(\log\frac{3}{2}-\frac{1}{3}\right)}$ is equal to (where $y_2(x)$ is $\frac{d^2y}{dx^2}$)

Space for Rough work

66. Let $f(x)=\frac{e^x-e^{-x}}{2}$ and if $f(g(x))=x$ then $\frac{1}{501}g\left(\frac{e^{1002}-1}{2e^{501}}\right)$ is equal to

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

67. Let $P_n(x)$ be a function satisfying $\sum_{n=0}^{\infty}h^nP_n(x)=\left(1-2hx+h^2\right)^{-1/2}$, $|x|\le 1, |h|<\frac{1}{3}$. Find $10P_3(10)$

68. If $\int_0^1\frac{dx}{(1+x)(2+x)\sqrt{x(1-x)}}=\frac{\pi A}{\sqrt{6}\left(\sqrt{3}+1\right)}$ then 13458.36A is equal to

69. One diagonal of a square is the portion of the line $\frac{x}{97}+\frac{y}{79}=1$ intercepted between the axes. p_1,p_2 are the lengths of the perpendiculars from the vertices of the other diagonal on the axis of y then $256.39(p_1+p_2)$ is

Space for Rough work